NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Data Science Fundamentals [SC3021]

Chapter 4: Data Preparation

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Chapter content

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Data Science Thinking vs. Computational Thinking

Data Science Ecosystem

Data preparation and data management

Flow of data

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Design considerations

Privacy

Security

Ethics

Psychology

Policies

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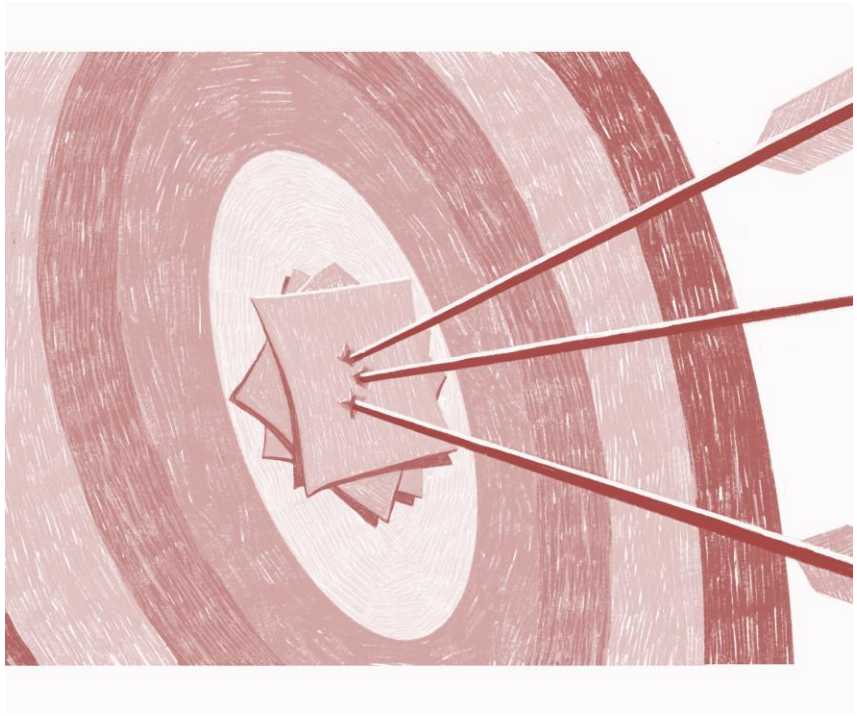
Chapter 4.1: Flow of data

At the end of this sub-chapter, you should be able to…

- … describe how data moves from raw to production data within organizations.
- … explain how different data preparation tasks may apply at different data science stages.
- … recognize and discuss which actions are involved in reaching different objectives.



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Data preparation definition

Data preparation is the process of **converting raw** data into a **form** that allows to extract **value** from data.

- Necessary to ensure that the data are …
  - … structured in a way that they can be input to the analysis stage (**structure**)
    - Many systems / algorithms used for analysis expect tabular data.
  - … of sufficient quality to ensure meaningful results (**content**).
    - Avoids “garbage-in-garbage-out”
    - Quality has many dimensions, e.g., completeness, correctness, timeliness, …

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Value from data

Data preparation is the process of **converting raw** data into a **form** that allows to extract **value** from data.

Value from data

Type of value

Near-term

Long-term

Value delivery

Indirect

Direct

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Type of value

- **Near-term**: clear questions to be answered with data
  Example: "Are people really shifting toward interacting with us through their mobile devices?"
- **Long-term**: exploratory questions with open answers that data can potentially support
  Example: "Can we understand how the move to mobile is affecting customers' purchasing patterns?

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Value delivery

- **Indirect**: Data provides value to an organization by influencing people's decisions or inspiring changes in processes.
  Example: risk modeling in the insurance industry.
- **Direct**: Data provides value to an organization by feeding automated systems.
  Example: Netflix's recommendation system.

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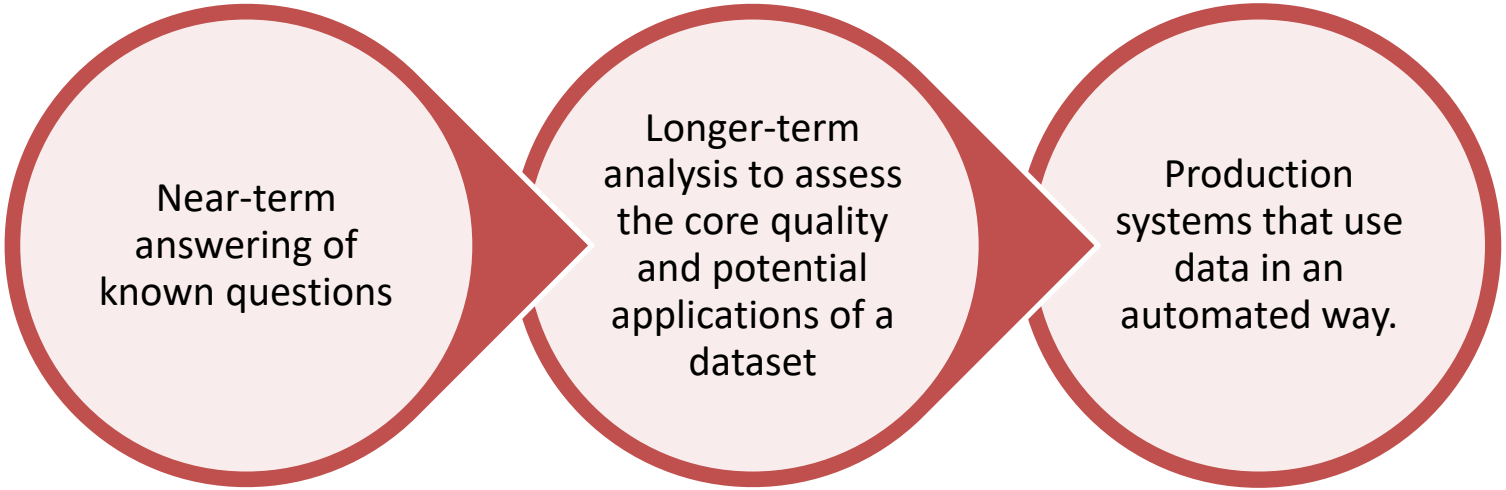
Progression of data projects



Movement of data

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Data movement through stages

Stage

Raw	Refined	Production

Actions

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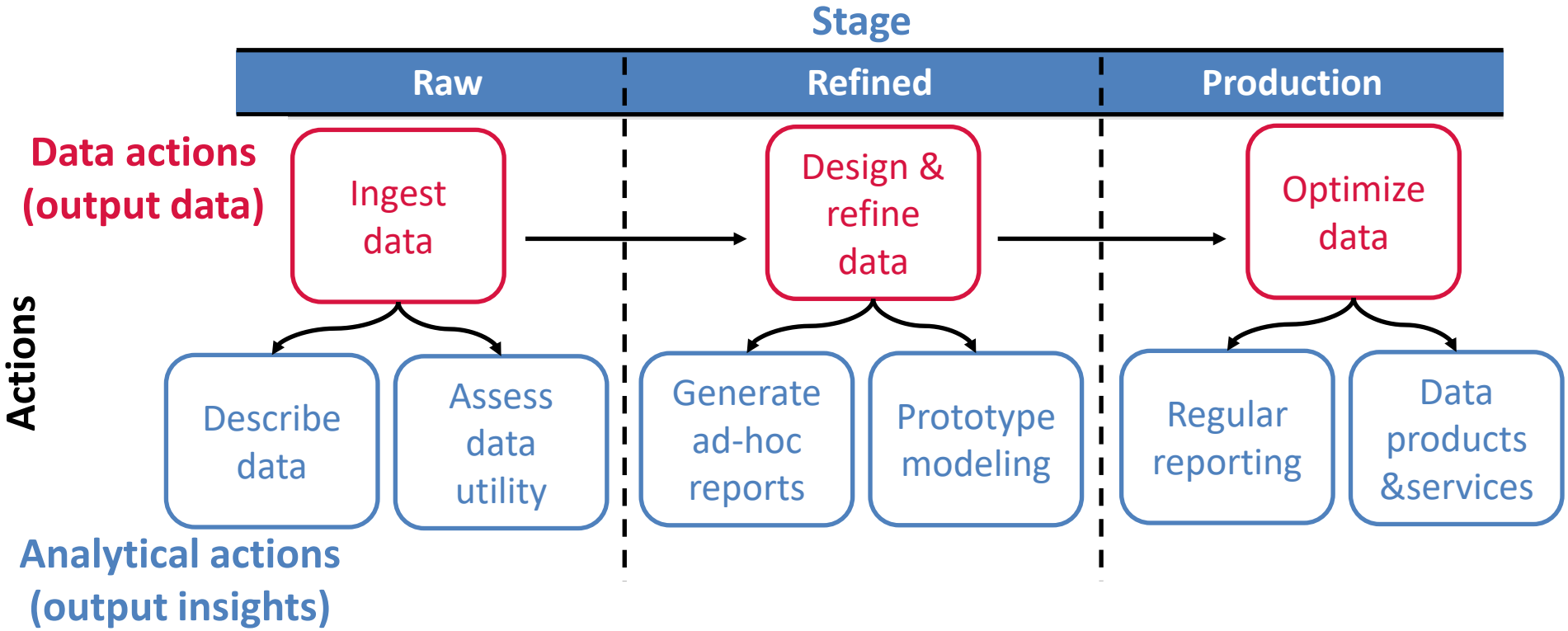
Data movement through stages



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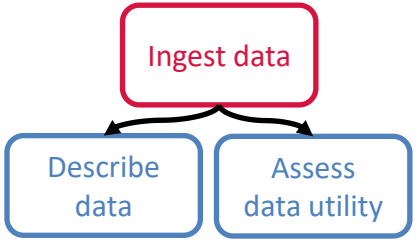
Raw Data Stage Actions



- **Ingest data** by accessing it and getting it in your sphere of control. Complexity highly varies (e.g., from downloading a file to using complex systems for real-time transfer of fine-granular data).
- When ingesting **(partially) unknown data**, additional **analytical actions** are typically necessary to **understand the data** and determine if it can / should be used.
  - **Describe data** by creating generic metadata (=data about data) to understand the characteristics of the data.
  - **Assess data utility** using the characteristics of the data, producing custom metadata.
  - Focus in both instances should be on understanding the **structure**, **granularity**, **accuracy**, **temporality**, and **scope** of the data.

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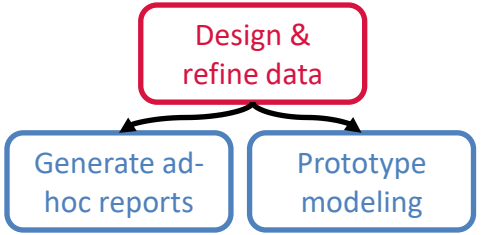
Refined Data Stage Actions



- **Design and refine data** to produce data that enables immediate application to a wide range of analyses. Involves a significant amount of transformation, in particular to address issues identified at the raw data stage.
- Refined data allow to **output insights** and information derived from the data.
  - **Generate ad-hoc reports** to answer specific retrospective questions using the data (e.g., “how many customers purchased item X last week”, “identify the key factors driving the customer trend of switching from desktop to mobile devices”)
  - **Prototype modelling** to understand the relevant factors that drive a behavior to make predictions about future behavior (e.g., “total sales next quarter”, “likelihood of contract renewal”)

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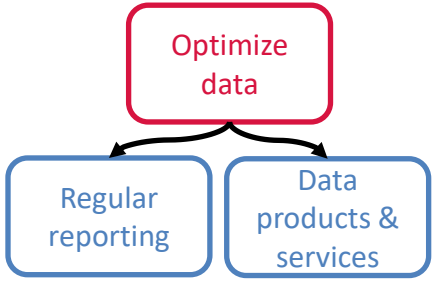
Production Data Stage Actions



- **Optimize data** to obtain data that simplify any additional downstream work for a highly specified intended use. The goal is to robustly and efficiently support a very narrow set of analyses.
- Production systems that operate in a largely **automated** way and with a well-defined level of **robustness** require further **tasks to operationalize analytics**.
  - **Regular reporting** through new (or updated) data that is processed in an ongoing manner, coping with variations / deviations from historical data.
  - **Data products and services** through generalized data preparation logic.

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Value from data (summary)

Data preparation is the process of **converting raw** data into a **form** that allows to extract **value** from data.

Indirect

Direct

Value from data

Near-term

Long-term

Ingest data

Describe data

Assess data utility

Design & refine data

Generate ad-hoc reports

Prototype modeling

Optimize data

Regular reporting

Data products & services

raw → refined → production

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Data conversion process

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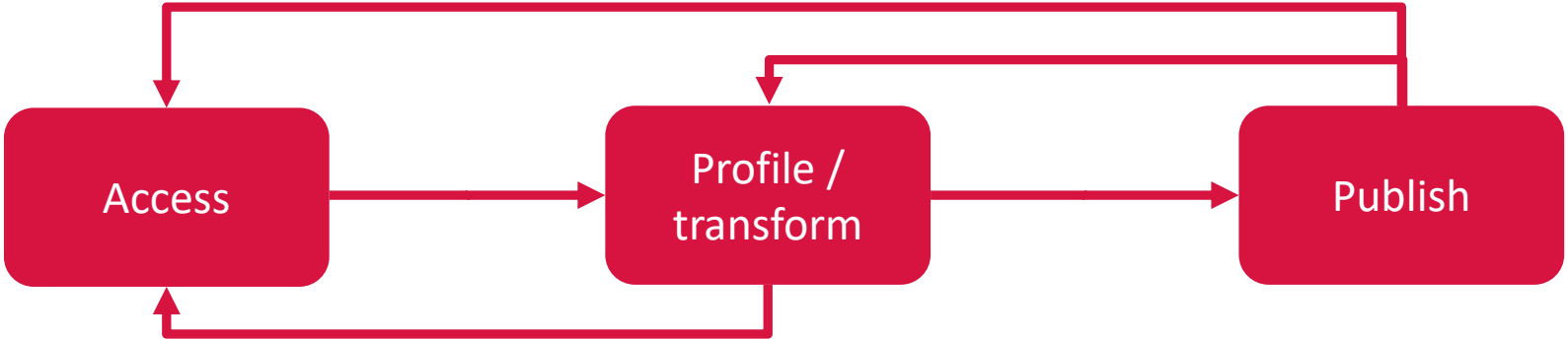

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Data preparation

Operations transforming data in multiple **stages** (cf Week 3):

- Data profiling
- Data structuring
- Data enriching
- Data cleaning

Profile / transform

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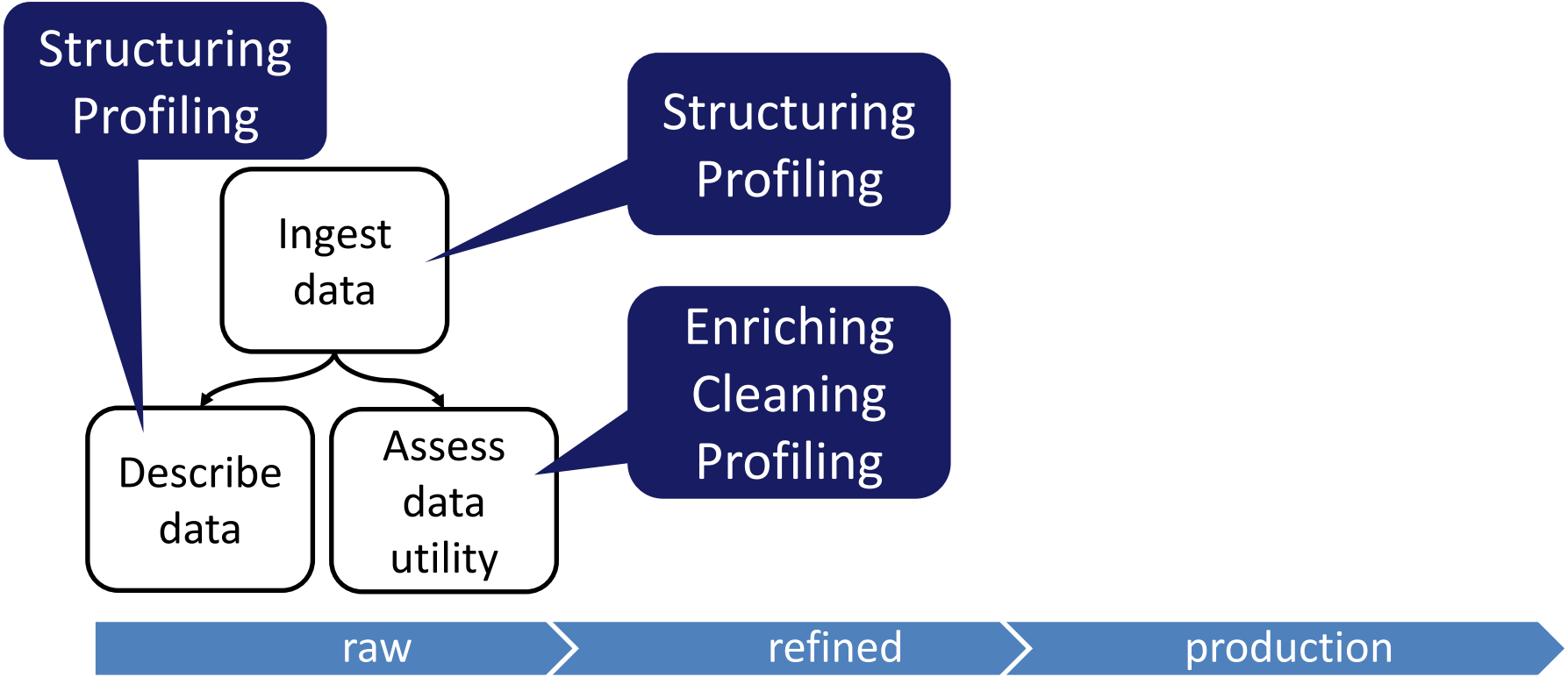
Data preparation along the flow of data



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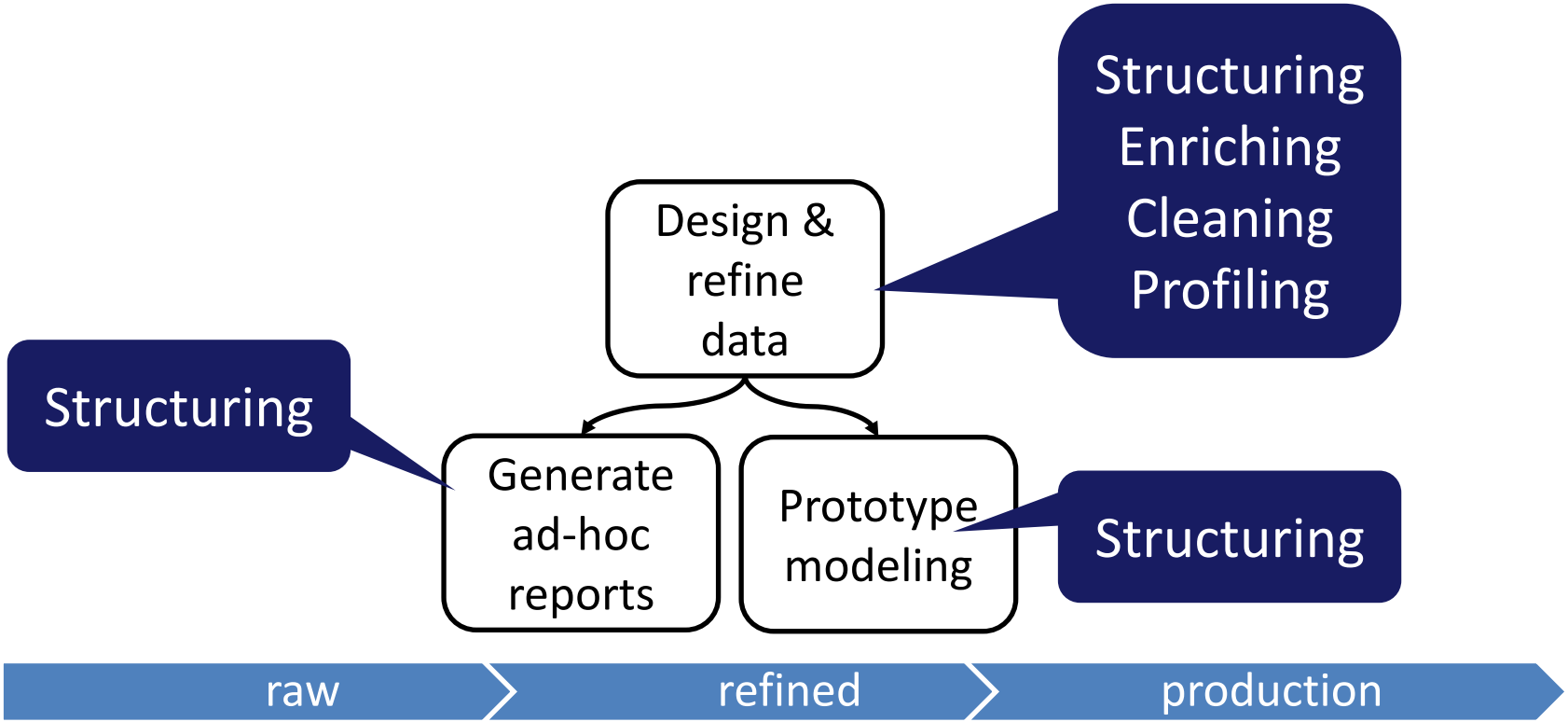

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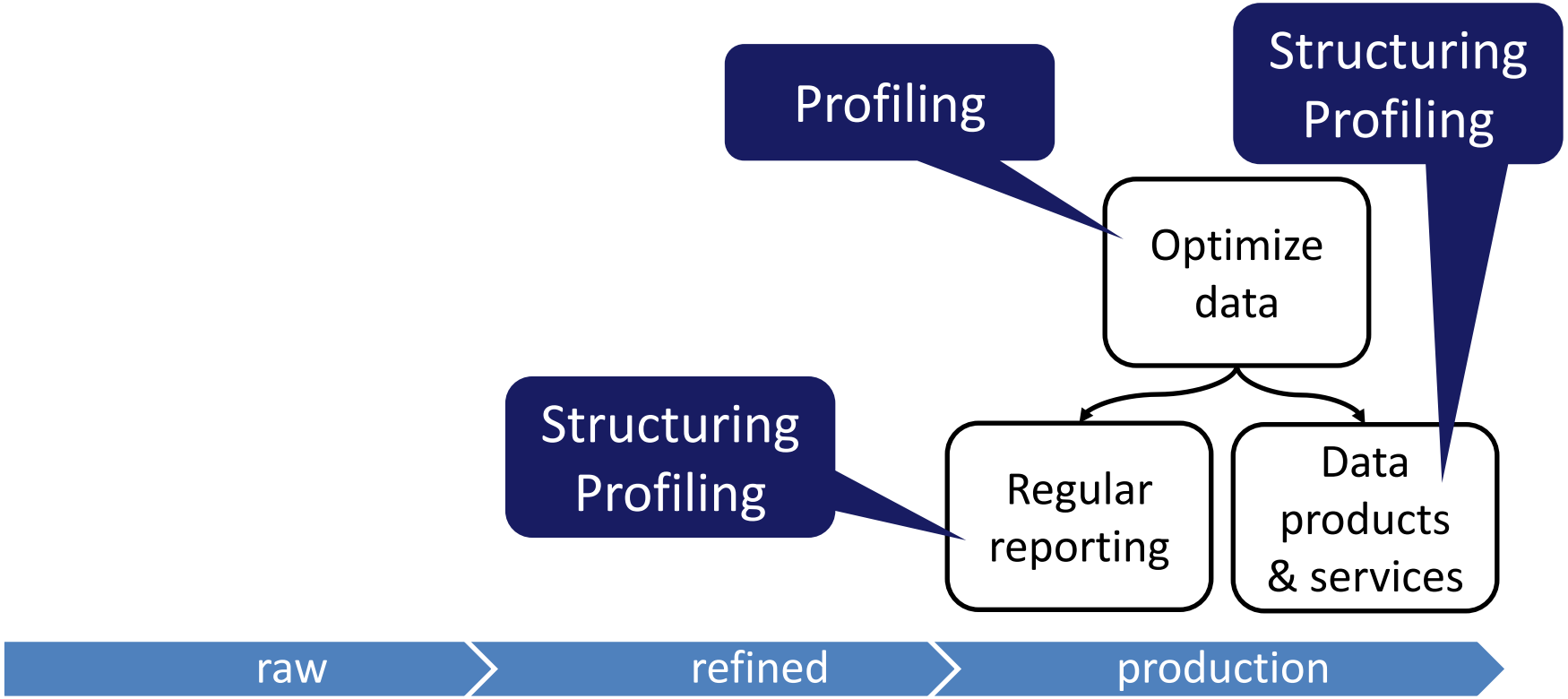

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Summary on flow of data

- Data evolves through stages (from raw over refined to production)
  - Not all applications go all the way (most frequent: refined)
  - Relates to value gained from the data.
- Data preparation is essential for data actions and analytical actions across all data flow stages.
  - Data preparation comprises several tasks (i.e., profiling, structuring, enriching, cleaning).
  - Relevance of different data preparation varies across data flow stages.

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You should now be able to…

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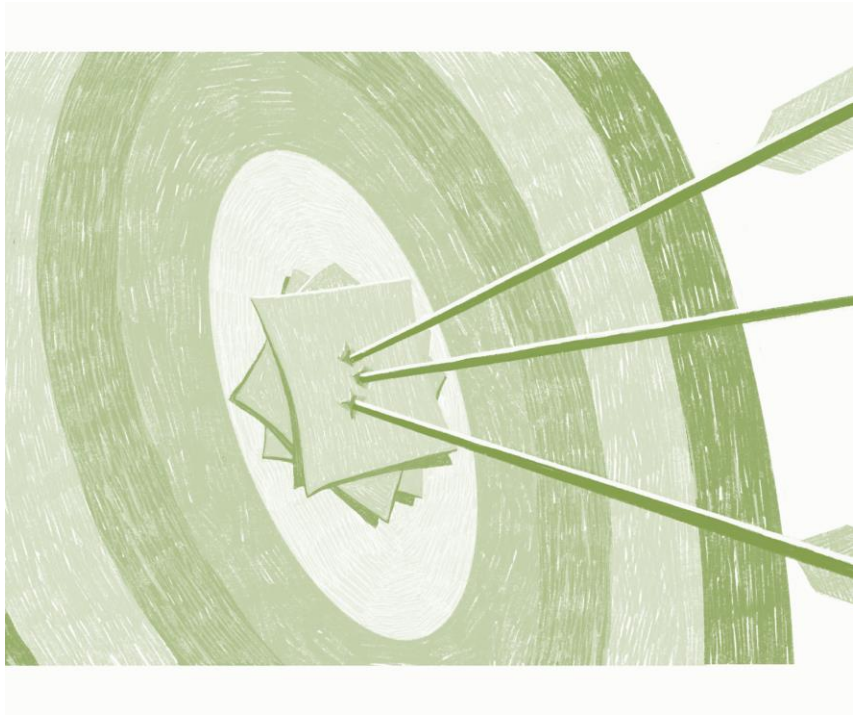

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References

- Joseph M. Hellerstein, Tye Rattenbury, Jeffrey Heer, Sean Kandel, Connor Carreras. Principles of Data Wrangling. O'Reilly Media. 2017

- Book available via NTU Library, content based on Chapters 2 and 3.

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